

Personality classification into 6 personality types according to the MBTI model



Teacher: Mohammed Al-Musa'idin

Student: Hassan Hussein Omar Al-Omari

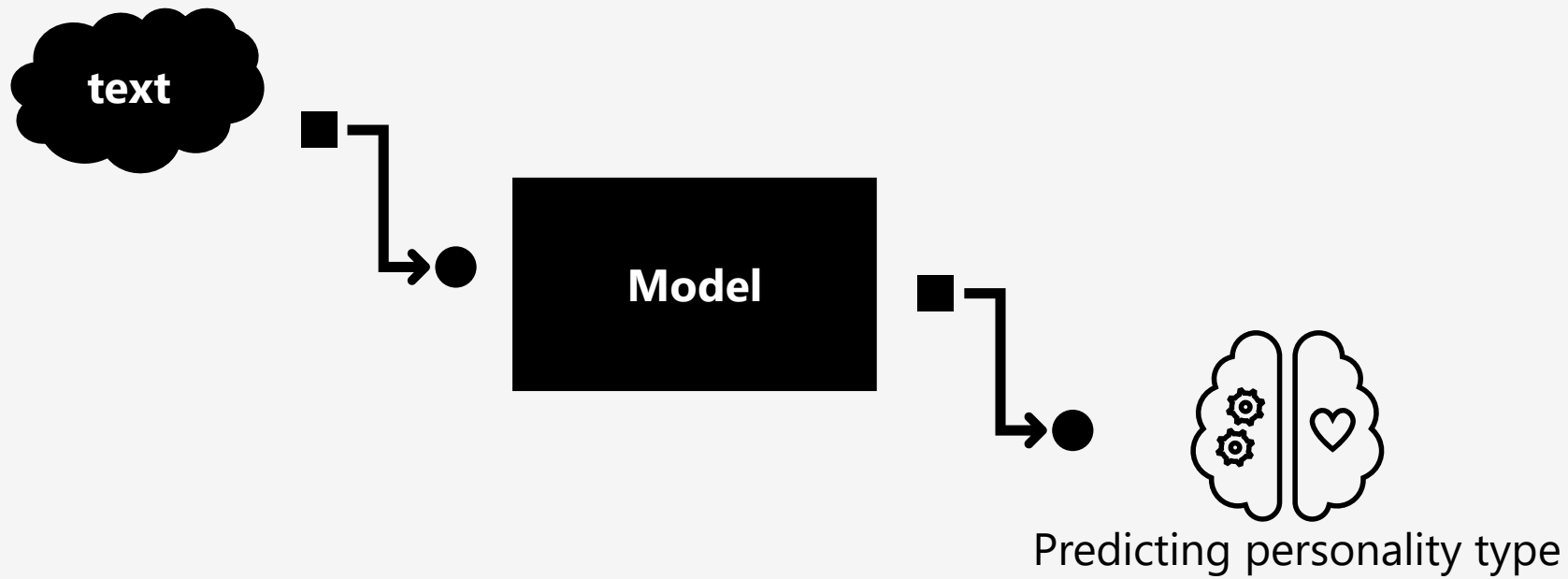
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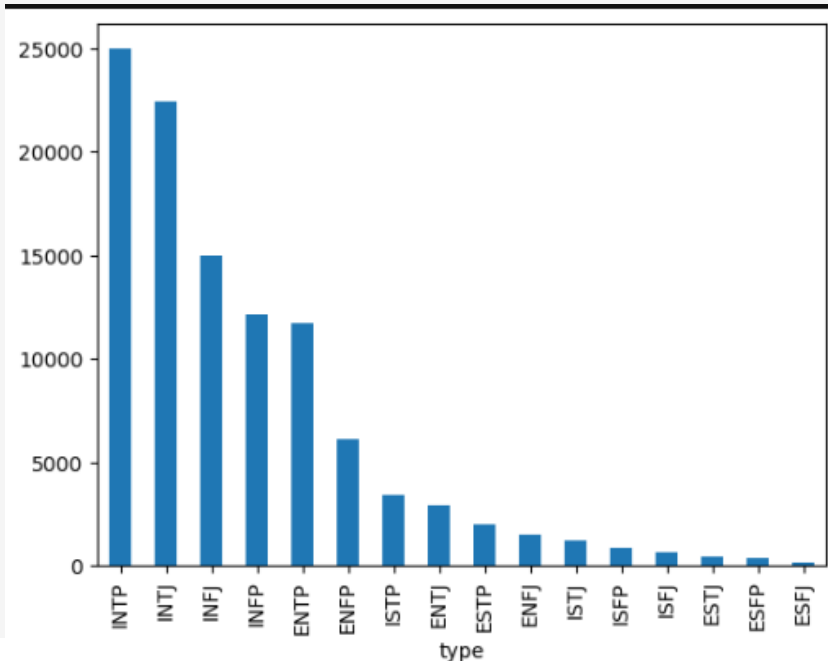
PROBLEM AND MOTIVE

It's difficult to know other people's personalities before getting to know them. It's illogical to go to someone you want to get to know and tell them to take a personality test. I created a model into which you enter a set of texts from a person, and the model predicted their personality type with an 86% accuracy rate. We can retrieve these texts from the person's posts on Facebook or Twitter, or even learn about an author's personality from their writings, or from someone's diaries, etc.



Dataset

This data from Kaggle consists of **106,067** rows. The first column, "**posts**," contains posts from individuals, each containing **500** words. The second column, "**type**," contains the label for the post. **This data has been pre-processed.**



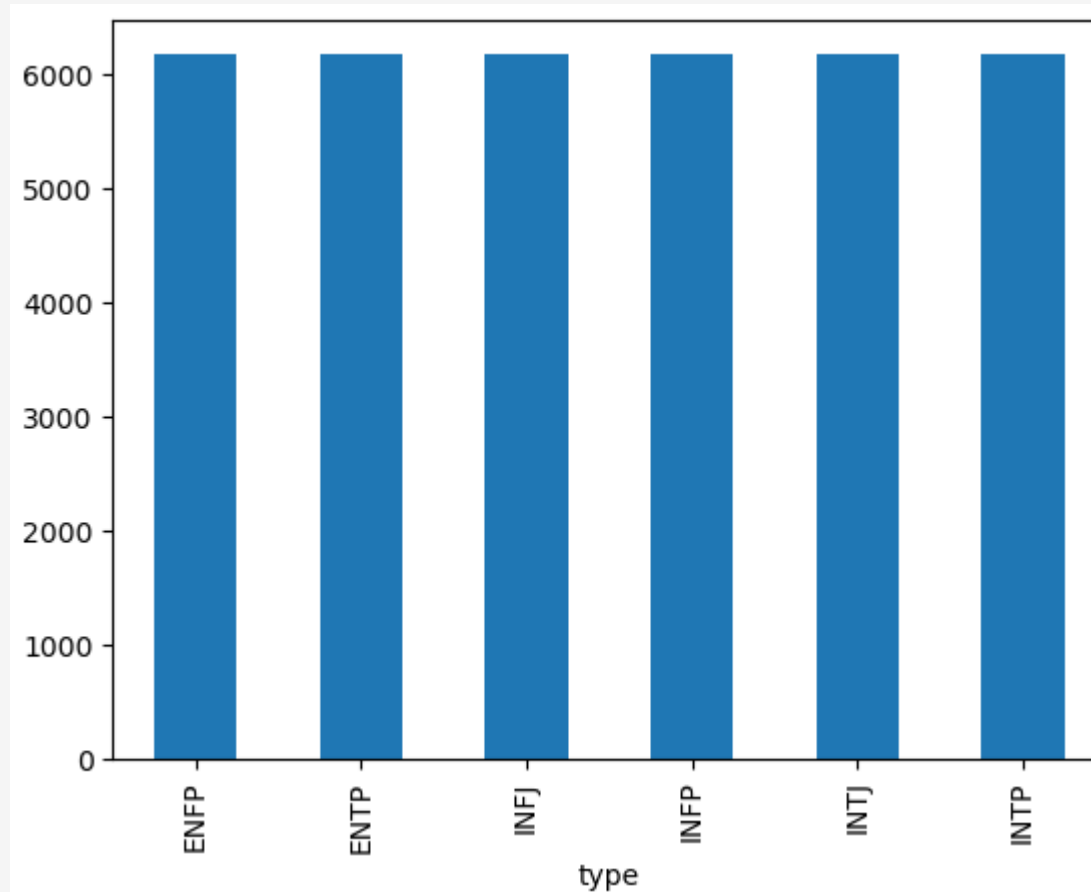
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[26]:
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	posts	type
0	idea would work figure thing physically expres...	ENFP
1	get field one example obviously advice always ...	ENFP
2	langston hughes speak souuuul love ti friend d...	ENFP
3	lay house kinda strange short term memory good...	ENFP
4	dislexia btw could pick hard word spell descri...	ENFP

This is the frequency of each character in the data set. However, it appears that the data is **unbalanced**, and if you enter it into the model, it will not produce correct results .

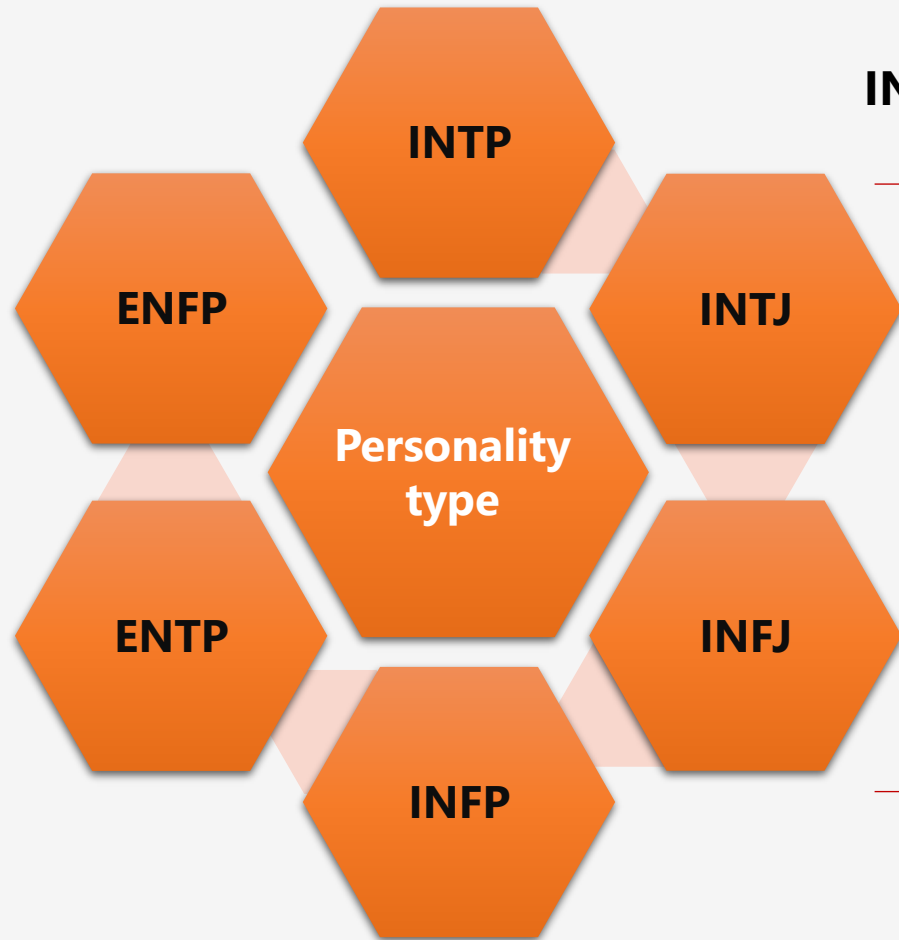
Dataset

Here I balanced the categories at their lowest frequency of 6,167 for each category in the data set.



Shape = (37002,2)

Data were balanced across 6 personality types



INTP: Analytical, curious, deep thinker, likes to understand things from the roots.

INTJ: Strategic, independent, plans intelligently, hates superficiality, focuses on big goals.

INFJ: Sensitive, calm, idealistic, loves to help others, deep in his feelings.

INFP: Creative, follows inner values, sensitive, loves artistic expression, flees from conflicts.

ENTP: Quick-witted, loves constructive debate, gets bored with routine, always has new ideas.

ENFP: Lively, enthusiastic, loves freedom, loves deep relationships, and is quick to get excited.

Technologies used

I used **linear SVC** to separate the classes and find the best "dividing line" between the classes in the data. Because my data is linear, I can separate each class with a line. This dividing line is used to be as far away as possible from the nearest points in each class, making the model robust to noise and outliers.

In order to be able to enter texts into the algorithm, they must be converted into a numerical representation. I used **TF_IDF**, which measures the importance of a certain word within a certain post, compared to all other posts. It converts the data into a format in which the rows are the number of each post, the columns are the unique words in all the data, and the values are the weights of the words relative to the post.

Results

I got 84% accuracy initially but through confusion_matrix I discovered that the model has a weakness in classifying some cases. I added some cases to the data to differentiate between them, and increased the accuracy to 86%.

before

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	ENFP	ENTP	INFJ	INFP	INTJ	INTP
ENFP	1086	28	48	53	28	9
ENTP	16	1085	26	12	35	69
INFJ	36	22	1018	70	41	17
INFP	47	21	66	1019	23	46
INTJ	24	23	35	20	1011	70
INTP	13	60	21	39	102	1062

: 0.8486690987704364



after

	ENFP	ENTP	INFJ	INFP	INTJ	INTP
ENFP	1083	28	44	58	30	10
ENTP	18	1073	26	14	39	70
INFJ	38	20	984	62	47	30
INFP	36	19	50	1029	27	34
INTJ	17	20	29	28	1076	67
INTP	8	43	20	29	74	1122

0.8603080248581464

Evaluation

precision_score = 0.86

recall_score = 0.859

f1_score = 0.86

Conclusion

The model is useful and effective in this case.

However, it needs more data and distinctive cases to be able to generalize to new data.

We can improve it in the future with more data so that it can identify more personality types.

Thank you.